Control work 1, and work on the mistakes

Limits, first principles and the five rules, in one paper — then an hour repairing what it finds.

LESSON 13–14 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 1 | P1 · CHAPTER 7 REVIEW

LESSON CLOCK

3 | 40' | 10' | 25' | 12'

Instructions	3 min
The paper	40 min
Self-mark	10 min
Rewrite and classify	25 min
Rule-selection drill	12 min

LEARNING OBJECTIVES

- Differentiate accurately under time.
- Choose the correct rule without being told which.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 71–74
Cambridge	Pure Mathematics 1, pp. 156–158
Workbook	Control paper A

01 Explanation

The paper — 25 marks, 40 minutes

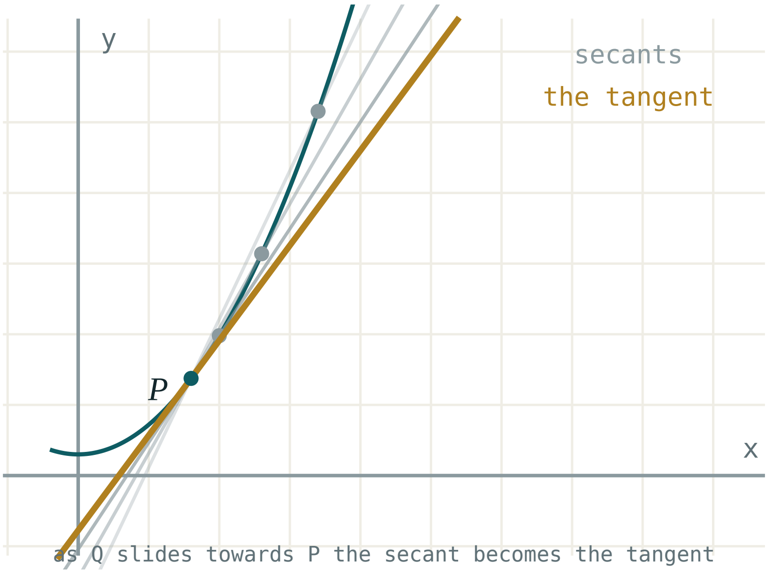
Q	TASK	MARKS	FROM
1	Evaluate $\lim_{x\to 3} \frac{x^2-9}{x-3}$ and $\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$	*4*	L3–4
2	Differentiate $f(x) = 2x^2 - 5x$ from first principles	*4*	L5–6
3	Differentiate $y = 3x^4 - \frac{2}{x} + \sqrt{x}$	*4*	L7–9
4	Differentiate $y = (2x+1)(x^2-3)$ by the product rule	*3*	L7–9
5	Differentiate $y = \frac{x}{x^2+1}$	*3*	L7–9
6	Differentiate $y = (3x^2-1)^5$ and $y = \sqrt{x^2+4}$	*4*	L10–12
7	A cube’s side grows at 0.2 cm/s. Find $\frac{dV}{dt}$ when the side is 5 cm	*3*	L10–12

Q2 IS MARKED ON THE DEFINITION

A correct answer obtained from the power rule scores **one** mark out of four. “From first principles” means the limit must appear.

The three columns

KIND	IN THIS TOPIC IT LOOKS LIKE	THE REPAIR
careless	a lost sign in the quotient rule, a dropped 2 in $\frac{du}{dx}$	check every answer by re-reading the rule
method	using the product rule where the chain rule was needed	the rule-selection drill below
knowledge	not knowing $(\sqrt{x})'$	learn the six standard derivatives



Question 2 in one picture — the secant, the limit, the tangent.

The rule-selection drill

Twelve expressions on the board. For each, the class calls out only the **rule**, not the answer, in five seconds:

EXPRESSION	RULE
$x^5 - 2x$	power and sum
$(x + 1)^7$	chain
$x^2(x + 1)$	product (or expand)
$\dfrac{x}{x + 1}$	quotient
$\dfrac{x^2 + x}{x}$	divide first, then power
$\sqrt{4x + 1}$	chain
$x\sqrt{x + 1}$	product then chain
$\dfrac{1}{(x - 2)^4}$	chain

ROWS 5 AND 8 ARE THE TRAPS

Row 5 needs no quotient rule at all; row 8 needs no quotient rule either, once it is written as $(x - 2)^{-4}$. Choosing the heavier rule costs time and invites sign errors.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: differentiate $f(x) = 2x^2 - 5x$ from first principles.**

1. $f(x + h) = 2(x + h)^2 - 5(x + h)$
2. $f(x + h) - f(x) = 4xh + 2h^2 - 5h$
3. $\frac{\cdots}{h} = 4x + 2h - 5$
4. $\lim_{h \to 0} = 4x - 5$

ANSWER

$f'(x) = 4x - 5$

EXAMPLE 2 **Model answer, Q5: differentiate $y = \frac{x}{x^2 + 1}$.**

1. $u = x,\ u' = 1;\ v = x^2 + 1,\ v' = 2x$
2. $\frac{1(x^2 + 1) - x(2x)}{(x^2 + 1)^2}$
3. $= \frac{1 - x^2}{(x^2 + 1)^2}$

ANSWER

$\frac{1 - x^2}{(x^2 + 1)^2}$

EXAMPLE 3 **A learner wrote $((3x^2 - 1)^5)' = 5(3x^2 - 1)^4$. Name the mistake.**

1. The inside derivative is missing.
   A method error.
2. $(3x^2 - 1)' = 6x$
3. Correct: $30x(3x^2 - 1)^4$.

ANSWER

Method error — the chain rule's second factor was omitted

03 Interactive model

Run the rule-selection drill before the rewrite, not after.

Which rule?

$y = (x^2 + 1)^6$

- power only
- chain
- product
- quotient

$y = x^3(x + 2)$

- chain
- product
- quotient
- none

$y = \frac{x^3 + x}{x}$

- quotient
- divide first
- chain
- product

$y = \frac{1}{(2x + 1)^3}$

- quotient
- chain on $(2x + 1)^{-3}$
- product
- power only

$y = \sqrt{x^2 + x}$

- chain
- product
- quotient
- power only

$((3x^2 - 1)^5)'$ is:

- $5(3x^2 - 1)^4$
- $30x(3x^2 - 1)^4$
- $15x(3x^2 - 1)^4$
- $6x(3x^2 - 1)^4$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Careless error	E’tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo‘shlig‘i	Пробел в знаниях
Mark scheme	Baholash sxemasi	Схема оценивания
Working	Yechim yozuvi	Ход решения
Correction	Tuzatish	Исправление
Rule selection	Qoidani tanlash	Выбор правила

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

“From first principles” means:

- use the power rule
- use the limit definition
- use a calculator
- draw a graph

Forgetting $\frac{du}{dx}$ is:

- careless
- a method error
- a knowledge gap
- not an error

$\frac{x^2 + x}{x}$ is best differentiated by:

- the quotient rule
- dividing first
- the product rule
- first principles

The quotient rule numerator is:

- $u'v + uv'$
- $u'v - uv'$
- $uv' - u'v$
- $u'v'$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$

   ANSWER 6

2. First principles: $(2x^2)'$

   ANSWER $4x$

3. $(3x^4)'$

   ANSWER $12x^3$

4. $\left(\dfrac{2}{x}\right)'$

   ANSWER $-\dfrac{2}{x^2}$

5. $((2x + 1)(x^2 - 3))'$

   ANSWER $6x^2 + 2x - 6$

6. $\left(\dfrac{x}{x^2 + 1}\right)'$

   ANSWER $\dfrac{1 - x^2}{(x^2 + 1)^2}$

7. $((3x^2 - 1)^5)'$

   ANSWER $30x(3x^2 - 1)^4$

Medium · the standard question

7 PROBLEMS

1. $\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$

ANSWER $\frac{1}{4}$

2. First principles: $(2x^2 - 5x)'$

ANSWER $4x - 5$

3. $(3x^4 - \frac{2}{x} + \sqrt{x})'$

ANSWER $12x^3 + \frac{2}{x^2} + \frac{1}{2\sqrt{x}}$

4. $(\sqrt{x^2 + 4})'$

ANSWER $\frac{x}{\sqrt{x^2 + 4}}$

5. Cube side grows at 0.2 cm/s; $\frac{dV}{dt}$ at side 5

ANSWER 15 cm³/s

6. $(x^2(x + 1)^3)'$

ANSWER $x(x + 1)^2(5x + 2)$

7. $(\frac{2x - 1}{x + 3})'$

ANSWER $\frac{7}{(x + 3)^2}$

Hard · stretch and reason

7 PROBLEMS

1. $\lim_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4}$

ANSWER 3

2. First principles: $\left(\dfrac{1}{x}\right)'$

ANSWER $-\dfrac{1}{x^2}$

3. $\left(x\sqrt{x^2 + 1}\right)'$

ANSWER $\dfrac{2x^2 + 1}{\sqrt{x^2 + 1}}$

4. $\left(\dfrac{(x + 1)^2}{x - 1}\right)'$

ANSWER $\dfrac{(x + 1)(x - 3)}{(x - 1)^2}$

5. Find x where $y = (x^2 - 4)^3$ has a horizontal tangent

ANSWER $x = 0, \pm 2$

6. Sphere: $\dfrac{dr}{dt} = 0.5$ cm/s; find $\dfrac{dV}{dt}$ at $r = 4$

ANSWER 32π cm³/s

7. Find y'' for $y = (2x + 1)^4$

ANSWER $48(2x + 1)^2$

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. It is not optional.

1. Rewrite in full every control-work question that lost a mark, with the “I lost this because …” sentence.

2. Five problems from the section your knowledge column was heaviest in.
3. Differentiate $y = x^2\sqrt{2x + 1}$ and factorise the answer.
4. Learn the six standard derivatives and write them from memory.